Single–image georeferencing of a MicaSense frame over flat ground

Mapping image pixels to ground coordinates (E, N) from one capture — no second image, no panel, no GCP

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1. The idea

An orthomosaic carries a 6–parameter *affine* geotransform because orthorectification has already resampled the imagery onto a planar map grid, so pixel $\rightarrow$ ground is linear. A *raw* frame is a central (perspective) projection of the 3–D scene: a pixel defines a *ray*, not a point, and a ground coordinate is obtained only by intersecting that ray with a surface. For an agricultural field the surface is well approximated by a plane, and then the pixel $\leftrightarrow$ ground map of one frame is an *exact* 3×3 projective homography (8 parameters) — the projective generalisation of the ortho's affine. A tilt away from nadir contributes the off–diagonal (keystone) terms that an affine cannot represent.

2. Camera model from the image metadata

Every parameter below is read directly from the file's XMP/EXIF.

Intrinsics (pixels). With ρ = `FocalPlaneXResolution` ($\mathrm{px\,mm^{-1}}$), focal length $f = f_{\mathrm{mm}}\rho$ and principal point $(c_x, c_y) = (\mathrm{pp}_x\rho,\ \mathrm{pp}_y\rho)$, giving

$$\mathbf{K} = \begin{bmatrix} f & 0 & c_x \\ 0 & f & c_y \\ 0 & 0 & 1 \end{bmatrix}.$$

We use the OpenCV camera frame: x right, y down, z along the optical axis (downward at nadir).

Lens distortion. Brown model, radial k_1, k_2, k_3 and tangential p_1, p_2 (`Camera:PerspectiveDistortion`, in that order).

Exterior orientation. The projection centre is $\mathbf{C} = (E_0, N_0, h)^\top$, where (E_0, N_0) is the GPS longitude/latitude projected into the local UTM zone and h the GPS altitude (orthometric). The orientation is the rotation $\mathbf{R}$ that takes camera axes into the world East–North–Up (ENU) frame, built from the EXIF Yaw/Pitch/Roll (ψ, θ, φ):

$$\mathbf{R} \;=\; \mathbf{C}_{en}\,\mathbf{C}_{nb}(\psi, \theta, \varphi)\,\mathbf{R}_{cb}, \qquad \mathbf{R}_{cb} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{C}_{en} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}.$$

Here $\mathbf{R}_{cb}$ is the fixed look–down camera$\rightarrow$body map, $\mathbf{C}_{en}$ swaps the navigation NED frame to ENU, and $\mathbf{C}_{nb} = \mathbf{R}_z(\psi)\mathbf{R}_y(\theta)\mathbf{R}_x(\varphi)$ is the body$\rightarrow$NED rotation (Bäumker convention, ψ yaw, θ pitch, φ roll).

3. From a pixel to ground coordinates (E, N)

For a pixel (u, v):

(i) Normalise: $x_d = (u - c_x)/f, \quad y_d = (v - c_y)/f.$

(ii) Undistort — recover the ideal (x, y) by inverting the Brown model, with $r^2 = x^2 + y^2$,

$$x_d = x\big(1 + k_1 r^2 + k_2 r^4 + k_3 r^6\big) + 2p_1 xy + p_2(r^2 + 2x^2),$$

$$y_d = y\big(1 + k_1 r^2 + k_2 r^4 + k_3 r^6\big) + p_1(r^2 + 2y^2) + 2p_2 xy,$$

solved by a few fixed–point iterations (MicaSense distortion is sub–pixel except at the edges).

(iii) Form the world ray: the camera–frame direction $\mathbf{d}_c = (x, y, 1)^\top$ rotates to $\mathbf{d} = \mathbf{R}\,\mathbf{d}_c$.

(iv) Intersect the ground plane $Z = Z_g$ (with Z_g the field elevation). The ray is $\mathbf{P}(t) = \mathbf{C} + t\,\mathbf{d}$; the plane fixes the scale,

$$\boxed{t = \frac{Z_g - C_z}{d_z}, \qquad \begin{bmatrix} E \\ N \end{bmatrix} = \begin{bmatrix} C_x \\ C_y \end{bmatrix} + t \begin{bmatrix} d_x \\ d_y \end{bmatrix}.}$$

That pair (E, N) is the ground coordinate of the pixel, in UTM.

Equivalent closed form (the matrix). For a plane the entire map is a single homography. Place the local ground origin at the nadir point (plane $Z = 0$, camera at $(0, 0, H_A)$ with $H_A = h - Z_g$). Then

$$s\,(u, v, 1)^\top = \mathbf{K}\,[\,\mathbf{r}_1 \;\; \mathbf{r}_2 \;\; \mathbf{t}\,]\,(E, N, 1)^\top, \qquad \mathbf{t} = -H_A\,\mathbf{r}_3,$$

where $\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3$ are the columns of $\mathbf{R}^\top$ (world→camera). Inverting the 3×3 matrix gives (E, N) directly from (u, v); absolute coordinates are this plus (E_0, N_0). The relation is exact for a plane and applies to *undistorted* pixels (do step ii first).

4. What to expect

Because the model is geometrically exact for a planar surface, pixel ↔ ground round–trips to numerical zero. Two practical limits remain. **(a) Absolute position** inherits the onboard GNSS uncertainty — roughly 1–2 m for a standard fix (the EXIF `GPSXYAccuracy`/`GPSZAccuracy` report it), removable only with RTK/PPK or a ground control point. **(b) Canopy parallax:** at low flight height a differential crop canopy behaves like relief and shifts edge pixels by about $(h_{\text{canopy}}/H_A) \times$ (edge distance) — decimetre–scale for a 1 m canopy at 22 m AGL, vanishing toward nadir. A *uniform* field slope is harmless, since a homography handles any tilted plane.

5. Running it

Dependencies: `numpy`, `pyproj`, `Pillow`, `tifffile` (and `rasterio` only for GeoTIFF output).
Print the homography and self–checks:

```
python micasense_georef.py IMG_0010_4.tif --ground-elev 481.5
```

Write a north–up GeoTIFF (UTM zone auto–selected from the GPS):

```
python micasense_georef.py IMG_0010_4.tif --ground-elev 481.5 --gsd 0.012 --out out.tif
```

`--ground-elev` is the field elevation in metres (from a DEM; it sets the footprint's scale and position), and `--gsd` the output pixel size in metres. Run the script per band *after* band–to–band alignment to build a georeferenced multispectral stack.